Requirement Document: N8N Hands-On Beginner Course

Topic 12: API & Data Storage Integration

1. Document Control

Field	Value
Document Title	Requirement Document: Topic 12 - API & Data Storage Integration
Course	N8N - Hands On Beginner Course
Target Audience	Trainee / Employee
Version	1.0
Status	Approved
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2. Objectives & Scope

2.1 Course Objective

The main objective of this learning module is to teach trainees and employees how to combine the HTTP Request node with Google Sheets to fetch external data and store it for later use. Trainees will learn the design pattern of fetching array-based JSON responses from a public API, resolving nested JSON structures, mapping those fields to a Google Sheet's columns, and understanding how n8n automatically handles multi-item loops.

2.2 Learning Scope

In Scope:

- Configuring an HTTP Request node to execute a GET request fetching a list of users (e.g., `https://jsonplaceholder.typicode.com/users`).
- Configuring a Google Sheets node (Append operation) to insert the fetched data.
- Resolving nested JSON elements (such as `company.name` inside a user profile) using expressions.
- Mapping dynamic data points (`id`, `name`, `email`, and `company.name`) to Google Sheet headers.

- Leveraging n8n's native looping mechanism to append all records in the fetched array to the spreadsheet.

Out of Scope:

- Handling complex deduplication logic or looking up existing spreadsheet rows before appending (focus is strictly on bulk data retrieval and appending).
- Dynamically formatting sheet cells or creating tables or columns via Google Sheets API.
- Connecting to third-party SQL databases (focus is strictly on Google Sheets as storage).

3. Target Audience Requirements (Pre-requisites)

To successfully complete this hands-on exercise, trainees/employees must have:

- Completed Topic 1 through Topic 11.
- Verified Google Sheets OAuth2 / Service Account credentials configured in n8n.
- A target Google Sheet created with columns: `ID`, `Name`, `Email`, `Company`.
- Internet access to fetch mock data from public REST APIs.

4. Functional Requirements

Requirement ID	Section	Requirement Description	Priority
REQ-001	API Fetch	The workflow shall include an HTTP Request node configured to perform a GET request fetching a list of records from `https://jsonplaceholder.typicode.com/users`.	**High**
REQ-002	Google Sheets Append	The workflow shall include a Google Sheets node configured with the Append Row operation to save the fetched records.	**High**
REQ-003	Nested Field Mapping	The Google Sheets node shall support extracting and mapping both flat and nested properties from the API response: • `id` to `ID` column • `name` to `Name` column • `email` to `Email` column • `company.name` (nested property) to `Company` column.	**High**
REQ-004	Multi-Item Looping	The workflow shall process all items returned in the API response array and append each as a separate row in the Google Sheet.	**High**

Requirement ID	Section	Requirement Description	Priority
REQ-005	Sheet Target Config	The Google Sheets node shall require specifying the target Spreadsheet ID and sheet name tab (`Sheet1`).	**High**
REQ-006	Complete Integration Flow	The workflow shall execute sequentially (Trigger -> HTTP Request -> Google Sheets) and write all retrieved items successfully.	**High**

5. Technical Requirements

ID	Technical Requirement	Details
TR-001	Automatic Iterative Execution	The n8n runtime must automatically loop downstream nodes when an upstream node outputs an array of multiple items.
TR-002	Nested Path Resolution	The expression engine must parse nested properties (dot notation: `{{ $json.company.name }}`) without throwing execution errors.
TR-003	Batch Write / Write Throttling	The Google Sheets node must successfully process multiple incoming rows (typically 10 rows from the API) in a batch or quick succession without hitting connection rate limits.

6. Stakeholders and Personas

Persona	Role	Need
Trainee / Employee	Student	Needs to learn the core data storage pattern of fetching data from external endpoints and writing it to a spreadsheet database for storage and reporting.

Persona	Role	Need
Instructor / Reviewer	Evaluation	Needs to verify that the student can combine HTTP and Google Sheets nodes, map nested properties, and successfully write multiple rows.

7. Assumptions & Constraints

- **Assumption 1 (ASM-001):** The public mock API returns an array of structured JSON profiles.
- **Assumption 2 (ASM-002):** The header row in the Google Sheet is row 1 and contains the exact column titles: `ID`, `Name`, `Email`, and `Company`.
- **Constraint 1 (CON-001):** Google Sheets API limits may restrict writing thousands of rows in simple loops, so this exercise is constrained to 10 mock records.